



University of the Witwatersrand
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Technical Report

HR Analytics Project
Data Cleaning and Workforce Metrics Analysis

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1 Introduction

According to Rogers and Jonker of IBM, "data cleaning is the process of identifying and correcting errors and inconsistencies in raw datasets to improve overall data quality". In any data-driven project, the quality of the input data directly influences the reliability of the outputs. Therefore, data cleaning serves as a foundational step in ensuring that subsequent analysis is both accurate and meaningful.

The objective of the data cleaning stage in this project was to transform the raw employee dataset into a structured, consistent, and reliable format suitable for analysis. This was achieved through the implementation of several data cleaning functions in Part A. These functions addressed issues such as missing values, inconsistent formatting, and logically invalid entries. Once the dataset was cleaned, it became suitable for Part B, where various key performance indicators were calculated. These metrics provide insights into workforce characteristics such as retention, turnover, performance, and demographic distribution.

This report is structured into two main sections. The first section explains the data cleaning techniques applied, highlighting both the problems identified and the approaches used to resolve them. The second section discusses the implementation of the metrics, including the logic used and the significance of each metric in supporting data-driven decision-making.

2 Data Cleaning

2.1 Remove Null Salaries

Some employee records contained missing salary values represented as NaN. Since salary is a numerical variable, missing values can lead to incorrect calculations or runtime errors during analysis. For example, calculating average salary would be inaccurate if such values are included.

To address this, each record was examined and the salary field (index 4) was extracted. The `math.isnan()` function was used to determine whether the value was missing. Records with valid salaries were retained, while those with missing values were removed. A new cleaned dataset was then returned.

2.2 Standardizing Departments

The department field contained inconsistent formatting, including uppercase, lowercase, and mixed-case entries. This inconsistency can result in incorrect grouping during analysis, where the same department is treated as multiple categories. To resolve this issue, each department name (index 1) was converted to lowercase using the `.lower()` method. The values were updated directly in the dataset, ensuring consistency without creating a new list.

2.3 Remove Invalid Performance Ratings

Performance ratings were expected to fall within the range of 0 to 5. However, some records contained values outside this range. Invalid ratings can distort performance-related metrics and lead to misleading conclusions. To address this, each rating (index 6) was checked to ensure it satisfied the condition $0 \leq \text{rating} \leq 5$. Records that did not meet this criterion were removed, and a new dataset containing only valid entries was returned.

2.4 Fix Hire Date Formatting

The dataset contained inconsistent date formats, including DD/MM/YYYY and YYYY-MM-DD, as well as extra whitespace. This inconsistency makes it difficult to perform comparisons and validation. To standardize the format, each date (index 9) was first cleaned using the `.strip()` method. If the date contained `"/"`, it was split into day, month, and year components and rearranged into the required format. Otherwise, the date was left unchanged.

2.5 Remove Invalid Hire Dates

Some hire dates were logically invalid or outside the acceptable range. The company was established in 2015, so any earlier dates are invalid, and dates beyond 2025 are considered future values. Each date was split into year, month, and day components and validated according to the following rules:

- Year must be between 2015 and 2025
- Month must be between 1 and 12
- Day must be valid for the month
- Leap year rules applied for February

Records failing any of these checks were removed.

3 Metrics

3.1 Unique Departments

The purpose of this function is to identify all distinct departments within the dataset. This is useful for understanding the organizational structure and for grouping data in further analysis.

The implementation uses a set, which automatically removes duplicates. Each department name is added to the set, resulting in a collection of unique departments.

3.2 Gender Distribution

This metric calculates the percentage distribution of genders within each department. It provides insight into workforce diversity and representation across departments. The implementation is performed in two stages. First, a nested dictionary is used to count the number of employees of each gender within each department. The outer dictionary represents departments, while the inner dictionary tracks gender counts. Once the counts are collected, the total number of employees in each department is calculated. Finally, the counts are converted into percentages that are rounded off to two decimals and then stored in the result dictionary. The formula that was used is as follows:

$$GenderDistributionPerDepartment(\%) = \frac{CountofGender}{TotalEmployeesinDepartment} \times 100$$

3.3 Average Age per Department

This metric calculates the average age of employees within each department, which can provide insights into workforce demographics. The approach groups ages by department using a dictionary where each key corresponds to a department and the value is a list of ages. After collecting all ages, the average for each department is calculated by summing the ages and dividing by the number of employees. The result is then rounded to two decimal places.

3.4 Retention Rate

The retention rate measures the proportion of employees who are still actively employed in the company. The implementation involves counting the number of employees with a status of "Active" and dividing this by the total number of employees in the dataset. The result is then multiplied by 100 to obtain a percentage:

$$RetentionRate = \frac{ActiveEmployees}{TotalEmployees} \times 100$$

3.5 Turnover Rate

This metric measures the proportion of employees who have resigned within each department. It helps identify departments with higher employee turnover. The implementation uses two dictionaries: one to track the total number of employees per department and another to track the number of employees who have resigned. The turnover rate is calculated for each department using the formula:.

$$TurnoverRate = \frac{ResignedEmployees}{TotalEmployees} \times 100$$

3.6 Average Salary by Age Range

This metric calculates the average salary of employees within a specified age range. It is useful for analysing compensation trends across different age groups. The function iterates through each employee record and selects only those whose age falls within the specified range. The salaries of these employees are summed, and the total is divided by the number of employees to calculate the average. If no employees fall within the range, the function returns 0 to avoid division by zero errors. The result is rounded to two decimal places. The result is rounded to two decimal places and stored in a dictionary.

3.7 Average Performance by Training Hours

This metric evaluates the relationship between training hours and employee performance across departments. The implementation filters employees based on whether their training hours fall within the specified range. Performance ratings for these employees are then grouped by department. For each department, the average performance rating is calculated by summing the ratings and dividing by the number of employees. If no employees in a department meet the criteria, a value of 0 is assigned. All results are rounded to two decimal places.

4 Conclusion

In this project, both data cleaning and metric functions played a critical role in transforming raw employee data into meaningful insights. The data cleaning stage ensured that inconsistencies, missing values, and invalid records were properly handled, thereby improving the reliability and accuracy of the dataset. The metrics developed in the second stage provided valuable insights into workforce characteristics, including departmental composition, employee retention, turnover, and performance trends. These metrics demonstrate how cleaned data can be used to support data-driven decision-making within an organization.

References

- [1] Jonker, A. and Rogers, J. (2025). *What is Data Cleaning?*. IBM Think. Available at: <https://www.ibm.com/think/topics/data-cleaning> [Accessed 17 April 2026].
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